

Charging infrastructure. For consumers to consider purchasing EVs, charging infrastructure needs to be developed in line with the current infrastructure of fuel pumps. Charging infrastructure typically includes both private and public charging stations. It is a vital part of the entire EV ecosystem. In fact, besides the upfront cost of the vehicle, availability of user-friendly charging infrastructure is the biggest factor that determines the success of a country's EV plan.

India currently has nearly 56,000 traditional fuel stations compared to just 222 community EV charging stations. Not only is developing the charging infrastructure ecosystem (in both urban and remote areas) imperative for EV adoption, charging stations need to be connected to the relevant electricity source—preferably from renewable sources such as wind or solar. Related policies also need to be revisited to assess whether charging stations will be regarded as electricity providers and, hence, governed by Electricity Act.

EESL has invited a bid for 2000 EV chargers for the second phase of its EV programme, which will see the roll out of 9500 EVs across different states.

Range. This refers to the distance an EV can run without requiring a recharge. EVs available in the market today have a relatively low range and are more suited for just city use. To realise their full potential akin to petrol vehicles, battery capacity needs significant improvements. However, with continued breakthroughs in battery research, EVs with a higher range are increasingly being developed.

While there are EVs in the market with a range as high as 400km to 500km, these largely fall in the high price segment. Cars within the more affordable segment typically average 200km to 300km per charge. Three new products being planned for launch by Mahindra in the next couple of years will probably have ranges of 200km to 350km.

Opportunity for the electronics industry

EVs are being designed with two motors per car (instead of one), which means requirement of many more motor controllers. Each vehicle has

EVs to drive huge demand for IGBTs

Yole Développement, a global market research company, forecasts that the EV market will help drive IGBT growth to US\$ 6 billion by 2018. IGBTs have long been at home in the switch power supply market, and these comprise the largest segment of the market for EV power systems. Vehicle electrification involves voltage levels between 12V and 400V, and requires high-power IGBT modules for drive and traction applications.

a charger and a battery management system onboard, both using power electronics. Moreover, EVs have electronic suspension, which means a bigger market for power electronic devices and components, including power management semiconductors and the like.

Major electronic components of EV systems include microcontrollers (MCUs), analogue gate drivers, power metal-oxide-semiconductor field-effect transistors (MOSFETs) and insulated-gate bipolar transistors (IGBTs), rectifiers, EEPROM memories, protection devices, voltage regulators and integrated circuits (ICs) for power management and battery monitoring systems.

Within the EV market, solutions to optimise power efficiency, ranging from high-power modules, MCUs and power semiconductors to sensors and discrete components are in high demand.

First area is the main inverter, which controls the electric motor to determine driving behaviour and captures kinetic energy released through regenerative braking, feeding recovered energy back to the battery.

Second is the DC-to-DC converter module, which supplies the 12V power system from the high-voltage battery.

Then comes the auxiliary inverter/converter, which supplies power on demand to systems such as air-conditioner, electronic power steering system, oil pump and cooling pump.

Fourth area is the battery management system, which controls the battery state during charging and discharging, to enable the longest possible battery life.

Fifth is the onboard charger unit, which allows the battery to be charged from a standard power outlet.

Complete solutions for EVs like for electric traction and energy recovery, power steering, automated manual

transmission, cooling system, pumps and power management are also in demand.

Innovative automotive applications like intelligent power switches for anti-lock brake systems, micro-electromechanical systems (MEMS), inertial sensors for automotive airbags and telematics microprocessors also see demand.

Wheels are another area of interest. Electrical wheels generate electricity, while hybrids may have thermoelectrics on exhaust systems and turbines, generating useful kilowatt-level power. Hence, its management will require electronics.

Power trains for EVs on land, water and air are evolving rapidly. By far, the biggest market will be for vehicles on land, particularly on-road and mainly for cars, followed by buses.

Moving forward

The desire for a longer driving range between charges, faster battery charging times, increasing electronics integration for infotainment, safety and security, and other applications may drive advances in areas beyond just power management, thus increasing the total electronics content in EVs.

For example, suspension will no longer be mechanical but an electrical active suspension generating electricity and using it, in part to manage itself, will deliver a far better ride and fuel economy. This, in turn, will involve more electronic components and devices.

Moreover, car entertainment systems with near-field communications (NFC) technology, vehicle access, in-vehicle networks and features related to secure connected mobility will enhance the electronics content in EVs. Therefore car-to-car communication, car-to-infrastructure communication, remote car management and broadcast reception solutions will be in demand. **EFY**